Newton - Euler Equation

Euler 1st law (translation) $\frac{d}{dt}{}^{I}\vec{P}_{a/o} = m_a {}^{I}\vec{a}_{a/o} = \vec{F}_{ex}$

$${}^{I}\vec{P}_{a/o} = m_a {}^{I}\vec{V}_{a/o} = m_a \frac{1}{m_a}\sum_i m_i {}^{I}\vec{V}_{i/o}$$

$$\frac{d}{dt}{}^{I}\vec{P}_{a/o} = m_a \frac{1}{m_a}\sum_i m_i {}^{I}\vec{a}_{i/o} = \sum_i {}^{I}\vec{F}_i = \sum_i \left[{}^{I}\vec{F}_{i,ex} + \sum_j {}^{I}\vec{F}_{i,j}\right] = \vec{F}_{ex}$$

Euler 2nd law (rotation) $\frac{d}{dt}{}^{I}\vec{h}_{s/o} = \sum_i {}^{I}\vec{r}_{i/a} \times {}^{I}\vec{F}_{i,ex} = {}^{I}\vec{M}_{a,ex}$

$$\begin{aligned}
\frac{d}{dt}{}^{I}\vec{h}_{s/o} &= \frac{d}{dt}\left[\sum_i m_i {}^{I}\vec{r}_{i/a} \times \vec{v}_{i/a}\right] \\
&= \sum_i m_i {}^{I}\vec{r}_{i/a} \times {}^{I}\vec{a}_{i/a} \\
&= \sum_i m_i {}^{I}\vec{r}_{i/a} \times \left[{}^{I}\vec{a}_{i/o} - {}^{I}\vec{a}_{a/o}\right] \\
&= \sum_i {}^{I}\vec{r}_{i/a} \times {}^{I}\vec{F}_i \\
&= \sum_i {}^{I}\vec{r}_{i/a} \times \left[{}^{I}\vec{F}_{i,ex} + \sum_j {}^{I}\vec{F}_{i,j}\right] \\
&= \sum_i {}^{I}\vec{r}_{i/a} \times {}^{I}\vec{F}_{i,ex} + \underbrace{\sum_i {}^{I}\vec{r}_{i/a} \times \sum_j \vec{F}_{i,j}}_{0 \text{ under internal moment assumption}} \\
&= \sum_i {}^{I}\vec{r}_{i/a} \times {}^{I}\vec{F}_{i,ex}
\end{aligned}$$

Newton Euler Equation : $({}^{I}\vec{M}_{a,ex})_{/B} = {}^{B}I_a \frac{d}{dt}\omega_{B/E} + \omega_{B/E} \times {}^{B}I_a \omega_{B/E}$

$$\begin{aligned}
{}^{I}\vec{M}_{a,ex} &= \frac{d}{dt}{}^{I}\vec{h}_{s/a} \\
&= \left(\frac{d}{dt}({}^{I}\vec{h}_{s/a})_{/B}\right)_{/E} + \omega_{B/E} \times {}^{I}h_{s/a} \\
&= \left[\frac{d}{dt}\left({}^{B}I_a \omega_{B/E}\right)\right]_{/E} + \omega_{B/E} \times \left({}^{B}I_a \omega_{B/E}\right)_{/E} \\
&= \left[{}^{B}I_a \frac{d}{dt}\omega_{B/E}\right]_{/E} + \left[\omega_{B/E} \times {}^{B}I_a \omega_{B/E}\right]_{/E} \\
&= \left[{}^{B}I_a \frac{d}{dt}\omega_{B/E} + \omega_{B/E} \times {}^{B}I_a \omega_{B/E}\right]_{/E}
\end{aligned}$$

Torque - Free Motion

when external torque = 0

1° ${}^{B}I_a \frac{d}{dt}\omega_{B/E} + \omega_{B/E} \times {}^{B}I_a \omega_{B/E} = 0$

$\frac{d}{dt}({}^{I}\vec{h}_{s/a})_{/E} = -\omega_{B/E} \times ({}^{I}\vec{h}_{s/a})_{/E}$

$\frac{d}{dt}({}^{I}\vec{h}_{s/a})_{/E}$ is perpendicular to $({}^{I}\vec{h}_{s/a})_{/E}$

$\|({}^{I}\vec{h}_{s/a})_{/E}\|$ is constant, $({}^{I}\vec{h}_{s/a})_{/E}$ lies on a "momentum sphere"

2° $T = \frac{1}{2} m_a \|{}^{I}\vec{V}_{a/o}\|^2 + \frac{1}{2}\omega_{B/E}{}^{T}\,{}^{B}I_a\,\omega_{B/E}$

$= \frac{1}{2} m_a \|{}^{I}\vec{V}_{a/o}\|^2 + \frac{1}{2}({}^{I}\vec{h}_{s/a})_{/B}{}^{T}\,{}^{B}I_a^{-1}\,({}^{I}\vec{h}_{s/a})_{/B}$ Conserved

$({}^{I}\vec{h}_{s/a})_{/B}{}^{T}\,{}^{B}I_a^{-1}\,({}^{I}\vec{h}_{s/a})$ is constant

$({}^{I}\vec{h}_{s/a})_{/B}$ lies on a "energy ellipsoid"

3° ${}^B I_G \frac{d}{dt} \omega_{B/B} + \omega_{B/B} \times {}^B I_G \omega_{B/B} = 0$

$\frac{d}{dt} \omega_{B/B} = {}^B I_G^{-1} \left[-\omega_{B/B} \times {}^B I_G \omega_{B/B} \right]$

let V be an eigenvector of ${}^B I_G$

when $\omega_{B/B}$ in the direction of V

$({}^I \vec{h}_{G/O})_{/B} = {}^B I_G \omega_{B/B}$ in the direction of V

$\frac{d}{dt} \omega_{B/B} = 0 \quad \frac{d}{dt} ({}^I \vec{h}_{G/O})_{/B} = 0$

along direction of eigenvectors of ${}^B I_G$, we get pure rotation

· Motion in Principle Axes Frame

$${}^B I_G = \begin{bmatrix} | & | & | \\ \hat{p}_1 & \hat{p}_2 & \hat{p}_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} I_1 & & \\ & I_2 & \\ & & I_3 \end{bmatrix} \begin{bmatrix} - & \hat{p}_1^T & - \\ - & \hat{p}_2^T & - \\ - & \hat{p}_3^T & - \end{bmatrix} = P \, {}^{BP} I_G \, P^T$$

${}^B I_G \frac{d}{dt} \omega_{B/B} + \omega_{B/B} \times {}^B I_G \omega_{B/B} = ({}^I \vec{M}_{G,ext})_{/B}$

$P \, {}^{BP} I_G \, P^T \frac{d}{dt} \omega_{B/B} + \omega_{B/B} \times P \, {}^{BP} I_G \, P^T \omega_{B/B} = ({}^I \vec{M}_{G,ext})_{/B}$

$P \, {}^{BP} I_G \frac{d}{dt} \omega_{B/BP} + \omega_{B/B} \times P \, {}^{BP} I_G \, \omega_{B/BP} = ({}^I \vec{M}_{G,ext})_{/B}$

${}^{BP} I_G \frac{d}{dt} \omega_{B/BP} + \omega_{B/BP} \times {}^{BP} I_G \omega_{B/BP} = ({}^I \vec{M}_{G,ext})_{/BP}$

$$\begin{bmatrix} I_1 & & \\ & I_2 & \\ & & I_3 \end{bmatrix} \begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\omega}_3 \end{bmatrix} + \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \times \begin{bmatrix} I_1 \omega_1 \\ I_2 \omega_2 \\ I_3 \omega_3 \end{bmatrix} = ({}^I \vec{M}_{G,ext})_{/BP}$$

angular velocity in principle axes frame

$$\begin{aligned} I_1 \dot{\omega}_1 + (I_3 - I_2) \omega_2 \omega_3 \\ I_2 \dot{\omega}_2 + (I_1 - I_3) \omega_1 \omega_3 \\ I_3 \dot{\omega}_3 + (I_2 - I_1) \omega_1 \omega_2 \end{aligned} = \begin{bmatrix} ({}^I \vec{M}_{G,ext})_{/BP} \end{bmatrix}$$

· Stability of Torque Free Motion

linearize about each axis and look at local stability

assume $I_1 < I_2 < I_3$

1° assume $\omega_2 \approx 0 \quad \omega_3 \approx 0 \quad \omega_1 \gg 0$

$I_1 \dot{\omega}_1 + (I_3 - I_2) \omega_2 \omega_3 = 0 \quad \Rightarrow I_1 \dot{\omega}_1 \approx 0$

$I_2 \dot{\omega}_2 + (I_1 - I_3) \omega_1 \omega_3 = 0 \quad \Rightarrow \dot{\omega}_2 = \underbrace{(I_3 - I_1)/I_2}_{\alpha_2 > 0} \omega_1 \omega_3$

$I_3 \dot{\omega}_3 + (I_2 - I_1) \omega_1 \omega_2 = 0 \quad \Rightarrow \dot{\omega}_3 = \underbrace{(I_1 - I_2)/I_3}_{\alpha_3 < 0} \omega_1 \omega_2$

$$\begin{bmatrix} \dot{\omega}_2 \\ \dot{\omega}_3 \end{bmatrix} = \begin{bmatrix} 0 & \alpha_2 \\ \alpha_3 & 0 \end{bmatrix} \begin{bmatrix} \omega_2 \\ \omega_3 \end{bmatrix}$$

$\Downarrow$

$\text{eig}(A) = \pm \sqrt{\alpha_2 \alpha_3}$

eigenvalues are pure imaginary

simple harmonic oscillation

2° assume $w_1 \approx 0$ $w_2 \approx 0$ $w_3 > 0$

$$I_1 \dot{w}_1 + (I_3 - I_2) w_2 w_3 = 0 \quad \Rightarrow \quad \dot{w}_1 = \underbrace{(I_2 - I_3)/I_1}_{\alpha_1 < 0} w_2 w_3$$

$$I_2 \dot{w}_2 + (I_1 - I_3) w_1 w_3 = 0 \quad \Rightarrow \quad I_2 \dot{w}_2 \approx 0$$

$$I_3 \dot{w}_3 + (I_2 - I_1) w_1 w_2 = 0 \quad \Rightarrow \quad \dot{w}_3 = \underbrace{(I_1 - I_2)/I_3}_{\alpha_3 < 0} w_2 w_1$$

$$\begin{bmatrix} \dot{w}_1 \\ \dot{w}_3 \end{bmatrix} = \begin{bmatrix} 0 & \alpha_1 \\ \alpha_3 & 0 \end{bmatrix} \begin{bmatrix} w_1 \\ w_3 \end{bmatrix} \quad \Rightarrow \quad \text{eig}(A) = \pm \sqrt{\alpha_1 \alpha_3}$$

eigenvalue are pure real one positive, one negative unstable

3° assume $w_1 \approx 0$ $w_2 \approx 0$ $w_3 > 0$

$$I_1 \dot{w}_1 + (I_3 - I_2) w_2 w_3 = 0 \quad \Rightarrow \quad \dot{w}_1 = \underbrace{(I_2 - I_3)/I_1}_{\alpha_1 < 0} w_3 w_2$$

$$I_2 \dot{w}_2 + (I_1 - I_3) w_1 w_3 = 0 \quad \Rightarrow \quad \dot{w}_2 = \underbrace{(I_3 - I_1)/I_2}_{\alpha_2 > 0} w_3 w_1$$

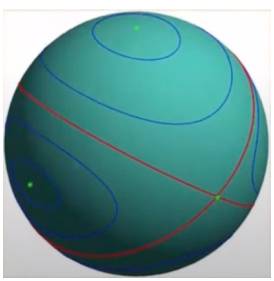
$$I_3 \dot{w}_3 + (I_2 - I_1) w_1 w_2 = 0 \quad \Rightarrow \quad I_3 \dot{w}_3 \approx 0$$

eigenvalues are pure imaginary simple harmonic oscillation

Spining around maximum / minimum axes is Lyapunov Stable

Spining around intermediate axis is unstable

Geometry on momentum sphere



green dots are equilibria

blue lines are bounded oscillatories

red line is the unstable one